FINAL RESEARCH REPORT · TASKS 1–3

Smooth source, unstructured head

A controlled flow-matching stress test

Egor Serov · Kirill Frolov · Daniil Koblov · Vasilii Lyamin

Question. Does the choice of a white-noise or smooth Gaussian-process source interact with the architecture of a conditional velocity network? We compare a Sundial-style adaptive-LayerNorm MLP with a small TSFlow-style S4 head, then add a PCA-whitened MLP control.

This report presents completed experiments on synthetic, history-normalized GP patches. It evaluates a narrow design hypothesis about source and head compatibility; it is not a pretrained Sundial or full TSFlow reproduction.

Evidence	Completed primary experiment
Design	2 sources × 2 heads, plus mandatory GP-source whitened MLP
Replications	5 matched optimization seeds per cell; 25 completed runs
Data	L=32, F=16; RBF lengthscale 8
Training	6,000 optimizer steps/run; batch 256; 64 Euler steps
Evaluation	8,192 held-out velocity pairs; 1,024 generated paths/run
Literal assignment rule	0/25 individual runs satisfy both two-sided 20% conditions

Geometry established. The matched-GP velocity has held-out top-two PCA energy 0.9784, above the assignment's 0.90 gate. The complete spectrum, source contrast and nonconstant-patch checks appear on page 3.

Measured GP-source comparison. Mean residual MSE is 0.04927 for the raw MLP and 0.05827 for S4. Relative to the raw MLP, whitening changes residual MSE by -18.6% and PC1 MSE by -8.8%. These comparisons concern the implemented models and matched training budget.

Smoothness is comparable. 25/25 runs satisfy the ±20% roughness condition. The separate velocity-error equality condition determines the literal pass outcome; failing it does not by itself mean generated paths are rough or the models are unusable.

Primary criterion outcome. No run passes the literal joint rule. This is reported as a failure of the specified pass criterion, not converted into a success using another normalization. Raw errors, normalized diagnostics, oracle noise and solver checks are shown separately. Any recommendation is consequently qualified.

Post-hoc finding (pp. 11–12). Measured directly, the head Jacobian shows the ill-conditioning the hypothesis predicts: with the GP source both heads leave the residual directions largely unlearned (MLP 16%, S4 32% of the exact response). It is harmless here, because those directions carry almost no variance in the source or the target, and the three Task 2 metrics cannot see it. This measurement goes beyond the planned metrics and does not change the outcome above.

Reading guide: scientific comparison (p. 2), data and models (pp. 3–4), required results (pp. 5–6), scale/noise/solver diagnostics (pp. 7–8), robustness and reproducibility (p. 9), optimization sensitivity (p. 10), post-hoc head Jacobian and supplementary analyses (pp. 11–12), recommendation (p. 13).

The missing comparison

TimeFlow in Sundial starts from white Gaussian noise and uses a small AdaLN MLP conditioned on a Transformer representation. Published forecasting results concern the pretrained system, including multi-patch prediction. GIFT-Eval evaluation uses 23 datasets, 97 configurations and 100 generated paths, with MASE and CRPS. These results support that complete system's forecasting performance; they do not establish whether its head is equally suitable for a smooth GP source. Liu et al., 2025.

TSFlow combines GP priors with a DiffWave-style residual backbone containing S4 temporal layers; it also investigates coupling and conditional-prior choices. Its eight-dataset evaluation uses nine-quantile CRPS estimates, 100 samples and five seeds. Prior ablations within that architecture demonstrate possible source benefits, but do not answer the head question when source and architecture both differ from Sundial. Kollovieh et al., 2025.

S4 supplies a structured state-space transformation along time. Its long-sequence modeling results motivate a temporal alternative to patchwise dense regression, but do not imply an advantage at F=16. A controlled short-horizon experiment is needed. Gu, Goel and Ré, 2022.

Aspect	TimeFlow / Sundial	TSFlow	This experiment
Source	White Gaussian noise	GP prior	White and matched-kernel GP
Velocity/transport head	AdaLN MLP	DiffWave-style S4	Small AdaLN MLP / 3-block S4
Conditioning	Pretrained Transformer	Forecasting backbone	Same tiny encoder design
Evidence	Full-system forecast scores	Forecast scores; prior ablations	Velocity-PC errors; path roughness
Question left open	GP source under MLP	Same source under matched heads	Source × head interaction

The initially disputed cell is **smooth source → smooth target with an unstructured MLP**, under independent coupling. White noise → smooth target is the published reference geometry, not the disputed cell. TSFlow is not a direct counterexample until heads and experimental conditions are matched.

Low-rank velocity covariance alone does not prove poor optimization. Flow-matching labels are random conditional targets; the learned field estimates their conditional mean. Whitening changes coordinates and the training loss, while S4 changes the function class and temporal bias. These interventions allow a local comparison, not automatic causal attribution. Lipman et al., 2023.

The standalone one-page critical synthesis is supplied as output/pdf/related-work.pdf; implementation revisions and adaptations are recorded in docs/model_provenance.md.

Verify the geometry before training

Each example is a joint GP draw containing 32 observed and 16 future points. The squared-exponential covariance has lengthscale 8, variance 1 and diagonal nugget 1e-06. Windows are independent. The primary split contains 32,768 training, 4,096 validation and 8,192 test windows.

Sundial’s released normalization is applied exactly: subtract the history mean and divide by its population standard deviation; replace a standard deviation ≤0.01 with 1. The future uses those same history statistics. No future value contributes to normalization.

Sources are independently drawn in normalized observation coordinates. “Matched GP” means the same kernel as the raw generator. After random context normalization, target and source marginal covariances are not identical; the report does not claim otherwise.

Held-out quantity	PC1 share	PC1+2 share	Within-patch SD	Roughness
Target patches	0.8447	0.9806	0.8030	0.1919
White-source velocity	0.7254	0.8504	1.3128	1.1530
GP-source velocity	0.8283	0.9784	0.9253	0.2207

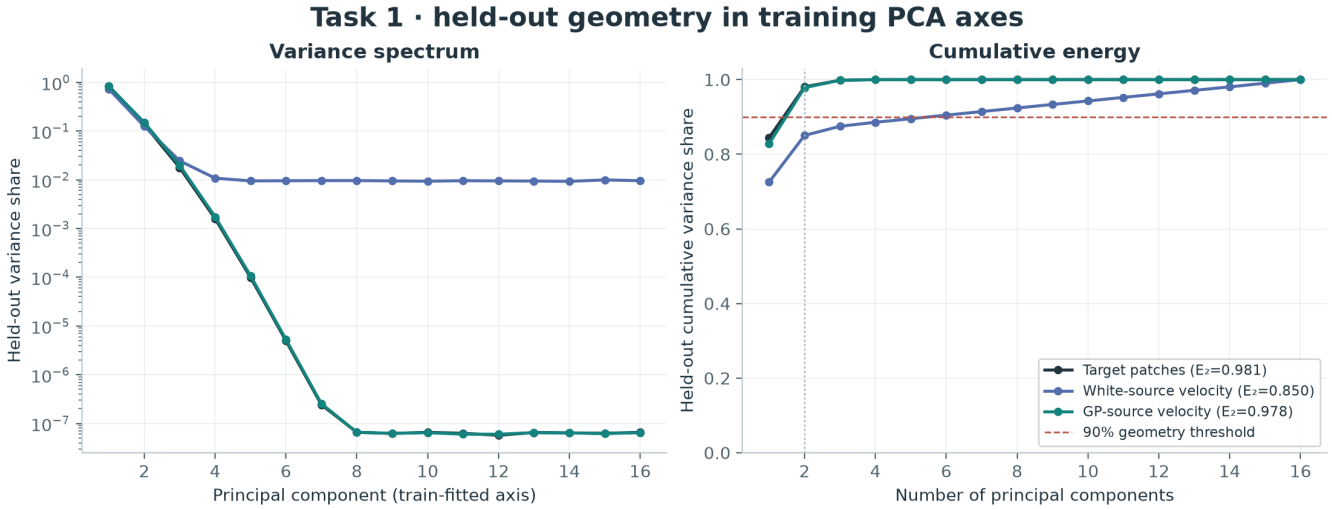


Figure 1. Test variance is projected into axes fitted only on training draws. The matched-GP velocity passes the geometry gate; white-source energy provides the numerical contrast. Nonzero within-patch variation rules out constant-series collapse.

Lengthscale 8 was chosen after a preliminary geometry pilot, before main model training. Primary data use a fresh seed. The supplementary lengthscale scan is disclosed as sensitivity analysis, not retroactive preregistration. The final test set did not determine the chosen lengthscale or checkpoints.

Matched capacity, explicit adaptations

The MLP adapts the released TimeFlow architecture: sinusoidal time embedding, AdaLN residual blocks and a conditional final readout. The S4 head uses three residual temporal blocks with genuine HiPPO-initialized normal-plus-low-rank S4, bidirectional temporal mixing, gated residual/skip branches and a small final readout.

Head	Parameters	Blocks	Width
White noise + MLP	65,464	3	56
White noise + S4	65,281	3	56
GP + whitened MLP	65,464	3	56

Both heads receive the same encoder architecture and identical initial encoder weights for each seed; each copy then trains jointly with its own head. Thus final encoder representations can differ. Capacity and training opportunity are controlled, but this is not a frozen, literally shared encoder comparison.

All runs use AdamW, learning rate 0.001, batch 256, gradient clipping 1 and 6,000 steps. SSM parameters retain no-weight-decay treatment. Validation selects checkpoints in each model's training coordinates; test labels are used only after selection. The whitened MLP therefore has a rescaled training/selection objective.

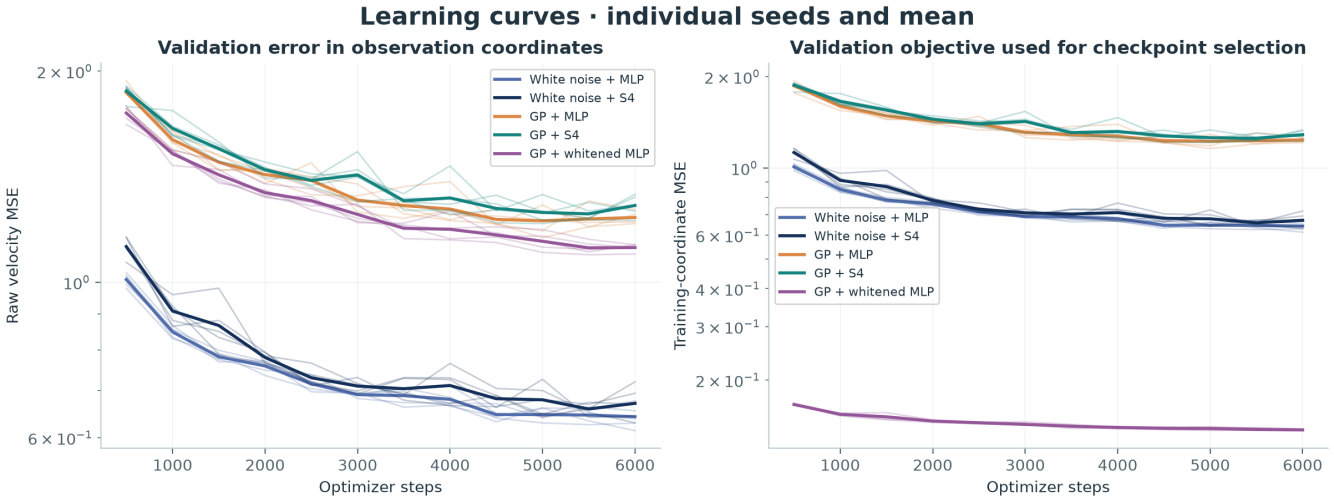


Figure 2. Thin curves show individual optimization seeds; thick curves show their means. Raw validation MSE is comparable in observation coordinates. The whitening objective has different coordinate weighting.

MPS adaptation. Complex HiPPO initialization runs once on CPU. Trainable S4 kernels use real state matrices, exact bilinear discretization and finite-horizon Krylov kernels on MPS. Numerical tests compare kernels and gradients against an independent complex reference. This implementation targets short patches, not long-sequence speed claims.

Objective disclosure. Released Sundial code predicts targets with its own weighted objective. This project reuses architecture but follows the assignment's explicit velocity target u=y−ε and independent interpolation. No pretrained weights or TimeBench pretraining are used.

Three metrics, five completed cells

Velocity error is projected into a source-specific PCA basis fitted to training velocities. PC1 measures the dominant direction; residual MSE averages per-component errors over PCs 3–F. PC2 is separately disclosed on page 7. Generated roughness is the mean absolute adjacent jump after the fixed Euler budget.

Cell	PC1 MSE	PCs 3–F MSE	Roughness	R / target	Passes
White noise + MLP	6.677 ± 0.084	0.1221 ± 0.0034	0.2005 ± 0.0068	1.058 ± 0.036	0/5
White noise + S4	7.321 ± 0.28	0.1249 ± 0.0077	0.198 ± 0.0089	1.045 ± 0.047	0/5
GP + MLP	14.93 ± 0.25	0.04927 ± 0.00077	0.1931 ± 0.007	1.018 ± 0.037	0/5
GP + S4	15.5 ± 0.49	0.05827 ± 0.0012	0.1969 ± 0.0084	1.039 ± 0.044	0/5
GP + whitened MLP	13.62 ± 0.15	0.0401 ± 0.00028	0.1926 ± 0.0039	1.016 ± 0.021	0/5

Mean ± sample SD across 5 optimization seeds. Target roughness = 0.189591. All generated paths and velocity errors are evaluated in the original normalized observation coordinates, including the inverse-mapped whitening control.

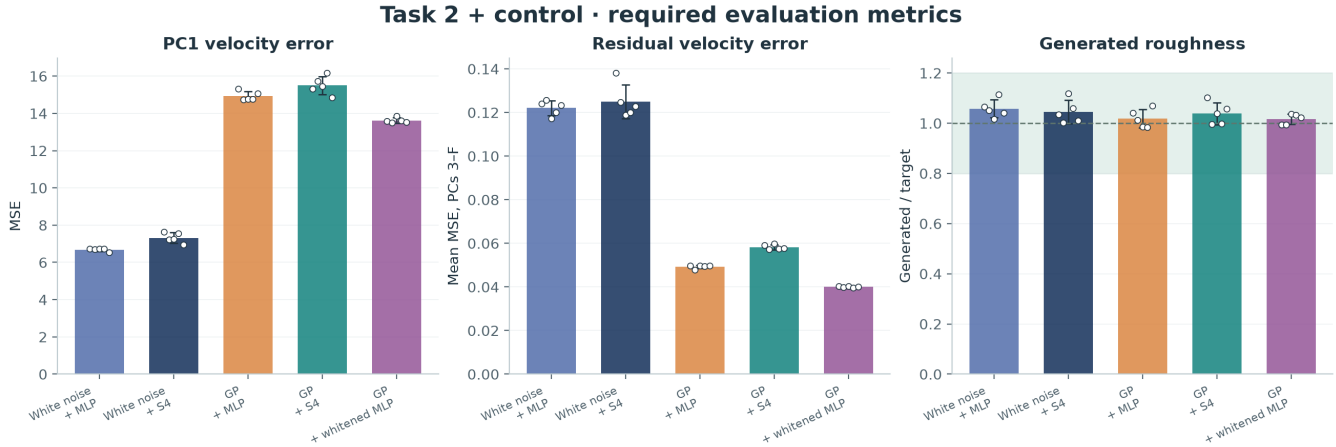


Figure 3. Points show individual seeds; summary uncertainty is seed SD. The green band marks the ±20% roughness interval. The first two panels show raw error magnitudes, not normalized substitutes.

Literal success rule: 0.8 ≤ residual MSE / PC1 MSE ≤ 1.2 AND 0.8 ≤ generated roughness / target roughness ≤ 1.2.

The rule is two-sided: residual error far below PC1 error also fails it. Because the GP spectrum is strongly anisotropic, this equality condition does not generally coincide with accurate regression. Pass counts are retained exactly; the following diagnostics examine this limitation without rewriting the criterion.

Paired 95% bootstrap intervals for all source/head and whitening contrasts are released in output/analysis/paired_seed_contrasts.csv. They resample matched seeds (10,000 resamples) and are conditional on this fixed dataset and hyperparameters. Five seeds provide limited interval resolution.

The same contexts in every cell

These are the first three held-out contexts (indices 0, 1 and 2) for seed 0, fixed by index rather than selected for appearance. The final eight observed history points are gray; the observed future draw is dashed; the generated future is solid. Each column uses the same vertical scale across all five models.

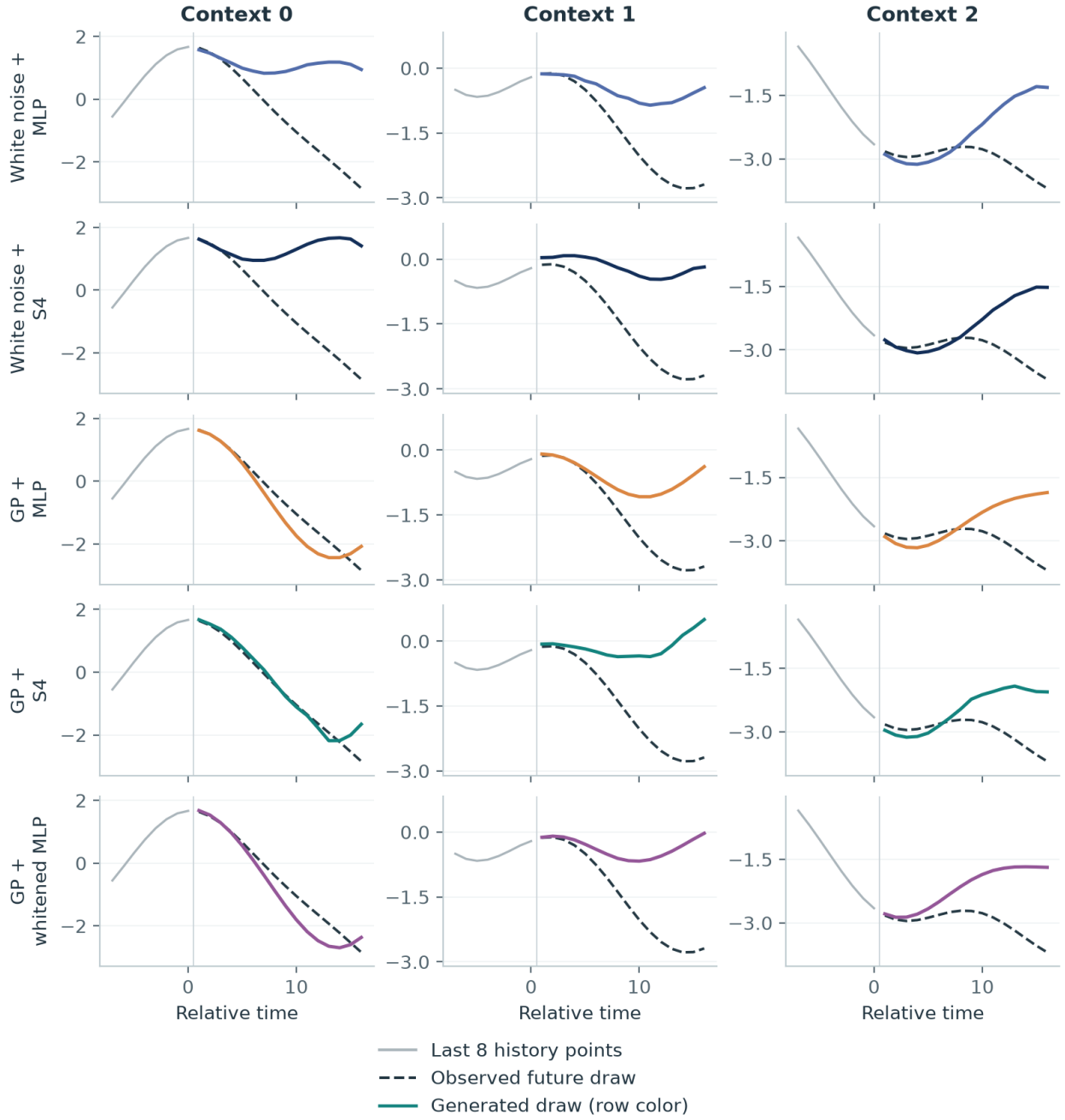


Figure 4. All four required cells and the whitening control are shown. The generated and observed futures are separate stochastic draws under the same context. Their pointwise difference is not treated as a unique deterministic forecast error.

The full six-context comparison is supplied as generated_paths_all_cells.pdf; the four-cell overview is generated_paths_2x2.pdf. The report uses fewer columns solely to keep axis labels legible. Complete arrays, histories and source draws are saved in results/main/samples/; plotted indices and provenance are recorded in path_figure_manifest.json.

06 · SCALE-SENSITIVE DIAGNOSTIC

Where the velocity error lies

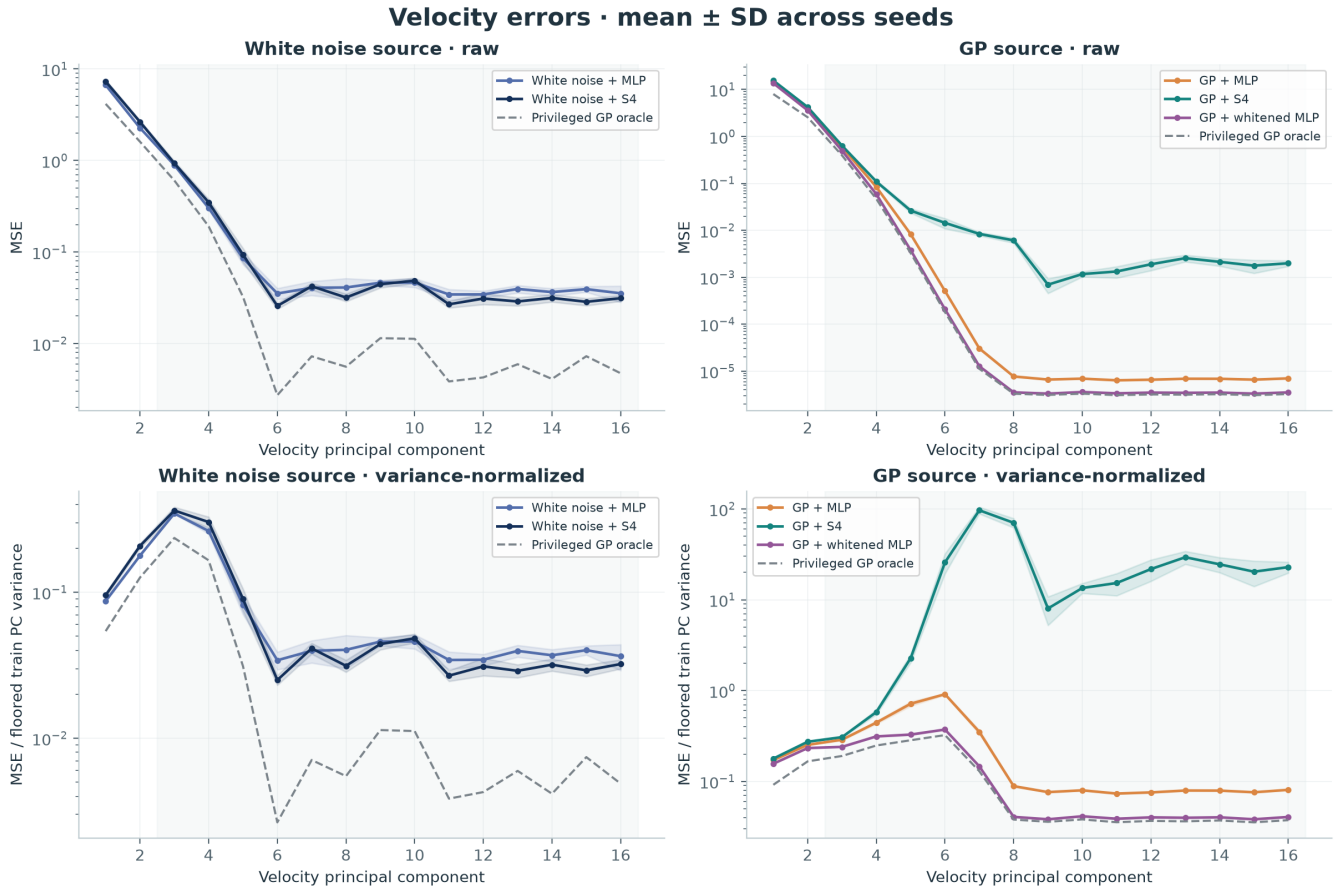


Figure 5. Mean raw and variance-normalized per-PC errors; shaded bands are seed SD. Gray dashed curves, when present, are the privileged GP oracle. The residual metric averages PCs 3–F; PC2 is retained visibly.

Cell	PC2 raw MSE	PC1 normalized	Residual normalized
White noise + MLP	2.269 ± 0.025	0.08712 ± 0.0011	0.08006 ± 0.0033
White noise + S4	2.644 ± 0.1	0.09553 ± 0.0036	0.08053 ± 0.0056
GP + MLP	3.895 ± 0.039	0.1717 ± 0.0029	0.2444 ± 0.0035
GP + S4	4.214 ± 0.023	0.1782 ± 0.0056	25.11 ± 3.1
GP + whitened MLP	3.591 ± 0.018	0.1566 ± 0.0017	0.1256 ± 0.00043

Normalized error divides each PC MSE by its training velocity variance, floored at $\max(10^{-6}\,\lambda_1, 10^{-8})$, before averaging residual PCs. The same floor defines whitening. This secondary measure reveals errors in small-variance directions but is not the assignment's primary raw MSE or a replacement pass criterion.

Actual GP floor: 8.69739e-05; 10/16 training-velocity directions fall below it. Their normalized errors use the floor rather than their smaller empirical variance. This regularization materially affects interpretation of the near-null tail and is shared with the whitening control.

Whitening uses an invertible linear map fitted on training velocities. Input patches, source, velocity and integrated outputs are transformed consistently; velocity targets are not incorrectly centered by a fixed mean. Whitening retains all coordinates and changes their scale—it does not by itself prove that F−2 dimensions were unused.

Separate regression noise from integration error

Conditional-flow labels $u=y-\varepsilon$ remain random given an interpolated state and context. Even the optimal conditional mean field can have nonzero label MSE. A Gaussian analytic diagnostic is computed using the complete raw history and its normalization statistics.

Privileged oracle	PC1 MSE	Residual MSE	Residual / PC1	Literal pass
White source	4.1542	0.064093	0.015429	False
GP source	7.9895	0.031854	0.003987	False

Information advantage. Learned heads see only normalized history, whereas the oracle sees the raw history and its location/scale. Its error is a privileged lower-bound diagnostic, not an attainable guarantee. The normalized target is not globally Gaussian; conditioning on raw history makes its affine Gaussian law exact.

Saved-checkpoint solver check	R ratio K=64	R ratio largest K	Endpoint RMSE / target RMS
GP + MLP	1.008	1.019	0.0179
GP + S4	1.063	1.078	0.0158
GP + whitened MLP	0.929	0.940	0.0154
White noise + MLP	1.083	1.098	0.0154
White noise + S4	1.042	1.055	0.0131

CPU check on the first 256 test histories, seed 0, using saved MPS-trained checkpoints. K=64 values differ from the primary 1,024-path estimates. Endpoint RMSE compares K=64 with K=256 under identical sources; primary scores are unchanged.

Oracle source	K	Expected R / exact	Conditional variance trace ratio	Directions <80% variance
white	32	0.9793	0.8935	0.866
white	64	0.9893	0.9451	0.824
white	128	0.9945	0.9721	0.804
white	256	0.9972	0.9860	0.770
gp	32	0.9836	0.9124	0.145
gp	64	0.9916	0.9551	0.122
gp	128	0.9958	0.9773	0.102
gp	256	0.9979	0.9886	0.067

Analytic covariance propagation can reveal suppressed conditional variance even when roughness looks accurate. This is a solver limitation of the near-singular Gaussian problem; accurate mean-path roughness alone does not establish distributional calibration.

What changes outside the primary setting

Lengthscale	Target E_2	White velocity E_2	GP velocity E_2
4	0.8629	0.6667	0.8486
6	0.9507	0.7889	0.9455
8	0.9795	0.8493	0.9776
10	0.9900	0.8806	0.9891
12	0.9943	0.9001	0.9937
16	0.9978	0.9204	0.9975
24	0.9993	0.9317	0.9991

Supplementary geometry scan, with independently seeded draws and training-only PCA. It does not select a trained winner. Long lengthscales can make white-source velocities nearly rank two as normalization inflates future variance, so smoothness alone does not guarantee the desired source contrast.

Lengthscale 6	Seeds	PC1 MSE	Residual MSE	R / target
White noise + MLP	3	6.591	0.1902	0.985
White noise + S4	3	6.561	0.1883	0.958
GP + MLP	3	17.47	0.1237	0.9411
GP + S4	3	17.26	0.1311	0.9341
GP + whitened MLP	3	16.54	0.1086	0.9434

GP-velocity test E_2=0.9461. Literal passes: 0/15; roughness passes: 15/15. Shortening the lengthscale does not resolve the raw equality criterion.

This second geometry is a labeled follow-up after the primary criterion audit. Its own configuration, split seed and runs are preserved. It does not overwrite the primary result or restore an architecture claim simply by changing the test.

Reproduction and leakage audit

Every trained run records seed, parameter count, validation-selected step, device, elapsed time and configuration hash. Test windows, sources and times are shared across matched comparisons; no test metric selects checkpoints. Checkpoints, training curves, PCA arrays and per-seed metrics remain available. The optional real-series extension is omitted; conclusions stay within the declared synthetic geometry.

Main commands: `python -m fm_stress.experiment train --config configs/main.json --output results/main --device mps`; then `python -m fm_stress.analyze --results results/main --output output/analysis --require-complete`. Geometry and oracle subcommands reproduce their separate records.

How sensitive is the ranking to training?

This is a separate post-primary sensitivity analysis: the three GP-source heads receive 3,000 extra optimizer steps at learning rate 0.0003, starting from their validation-selected checkpoints with Adam state reset. The original checkpoint remains eligible for selection. Main results and checkpoints are not overwritten.

The check first ran on seed 0. A validation-only improvement/ranking trigger expanded it to seeds 1 and 2, giving 3 matched seeds and 9 warm restarts. It is an adaptively extended diagnostic, not an independent preregistered confirmation or a replacement for the five-seed primary table.

GP-source head	Initial validation raw MSE	Selected validation raw MSE	Own objective improvement (%)
GP + MLP	1.189 ± 0.036	1.104 ± 0.026	7.141 ± 0.7
GP + S4	1.231 ± 0.019	1.125 ± 0.015	8.589 ± 1
GP + whitened MLP	1.113 ± 0.016	1.044 ± 0.022	3.097 ± 0.69

Mean ± seed SD. Raw validation MSE has common observation-space units. The final column measures within-model improvement in its selection objective; whitening uses a differently weighted objective, so those percentages are not a common cross-model score.

Validation raw-MSE ordering, lower is better: before restart, whitened MLP < MLP < S4; after restart, whitened MLP < MLP < S4. This ordering describes these three seeds and the bounded additional budget. It does not establish convergence or a universal architecture ranking.

Restarted head	Test PC1 MSE	Test residual MSE	R / target	Literal passes
MLP	13.74 ± 0.27	0.04626 ± 0.00071	1.009 ± 0.023	0/3
S4	14.09 ± 0.3	0.05693 ± 0.0018	1.038 ± 0.031	0/3
whitened MLP	12.71 ± 0.17	0.03855 ± 0.00012	1.005 ± 0.012	0/3

Test labels are evaluated only after the additional validation selection. These are supplemental outcomes of the changed training schedule. The same original-coordinate metrics and literal two-sided 20% rule are retained; an improved validation score is not relabeled as passing the assignment rule.

For exact per-seed before/after metrics, saved states, selected extra steps and learning curves, see results/main/optimization_sensitivity/. The validation-based expansion decision is recorded alongside the runs. Finite-budget sensitivity limits a mechanistic explanation based solely on the initial head comparison.

The head Jacobian

Run after the primary results; no primary flag changes. The exact, perturbation and window-length probes fixed their reading rules before their first runs; the per-PC breakdown is descriptive (scripts/probe_exact_jacobian.py, jacobian_per_pc.py, perturbation_recovery.py, jacobian_vs_window.py; outputs in results/diagnostics/).

The assignment locates the claim in "the Jacobian of the head". With a Gaussian source $N(0, C_s)$ and the exact Gaussian conditional target $N(m, C)$ given the raw history, the ideal velocity is affine in x_t, with Jacobian $J^* = U S^{-1}$, where $S = t^2C + (1-t)^2C_s$ and $U = tC - (1-t)C_s$. For every primary checkpoint we computed the full learned 16×16 Jacobian by automatic differentiation on 256 test windows at five times and compared it with J^* in the training-velocity PCA basis.

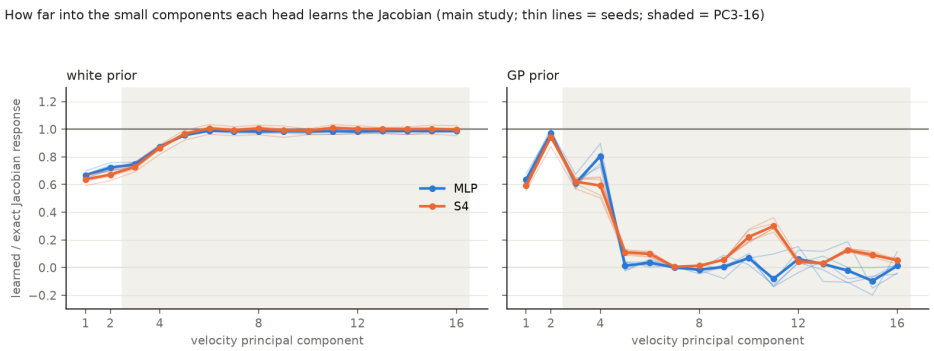
Source · head	Learned / exact norm, PCs 3–16	Cosine with J*, PCs 3–16	Relative error, PCs 3–16	Kick gain	CRPS change
White · MLP	0.98 [0.96–0.99]	0.99 [0.99–0.99]	0.13	0.23	+0.0000
White · S4	0.99 [0.95–1.02]	0.99 [0.99–0.99]	0.15	0.22	+0.0000
GP · MLP	0.16 [0.14–0.19]	0.21 [0.15–0.25]	0.99	1.82	+0.0003
GP · S4	0.32 [0.29–0.34]	0.34 [0.30–0.38]	0.93	1.87	+0.0004
GP · whitened MLP	—	—	4.33	26.79	+0.0697

Norm and cosine: mean over five seeds [range]. Kick gain: response at t=1 to a size-0.1 perturbation in PCs 3–16 injected at t=0.3; the exact field's gain is 0.16 (white) and 19.3 (GP). CRPS change: perturbed minus unperturbed ensemble CRPS.

The predicted ill-conditioning is present. With the white source both heads learn the residual block almost exactly. With the matched GP source both leave it largely unlearned: S4 has the lower residual-block error in every seed (MLP 0.979–0.993, S4 0.916–0.937), as the pre-declared rule required. Under the GP source the velocity variance along PC5 is $1.3 \cdot 10^{-4}$ of PC1 and falls to about $7.2 \cdot 10^{-8}$ along PCs 7–16, so gradient steps in those directions vanish.

S4 mitigates it only partly, and it is harmless here. The unlearned directions are those along which x_t never varies: generation starts from the same GP source, which already carries nearly the right spread there. A missing response therefore costs nothing in velocity MSE, generated roughness or CRPS. Neither head corrects an injected perturbation (relative error 0.99 for MLP and 0.97 for S4, against 0.99 when it is carried unchanged); S4 is lower in 5/5 seeds, with no forecast effect. The whitened MLP over-responds, amplifying the kick more than the exact field and worsening CRPS.

Window length. On the kernel-scan checkpoints (RBF, F = 16, 32, 64), S4's residual-block advantage does not grow with F; the pre-declared rule reads "refuted".



Learned response projected on the exact response, per input velocity PC (1 = exact, 0 = none); main-study checkpoints, mean over five times; thin lines are seeds.

Normalization, kernels and metrics

ETTh1-OT: global versus history normalization

The extension found S4 ahead of the MLP on ETTh1-OT with a GP source. That study scaled each channel globally, so the networks saw the absolute temperature. A pre-declared follow-up (docs/etth1_normalization_protocol.md, 24 fits) repeats the four cells with each window normalized by its own history, as in the synthetic task. Forecasts are mapped back to the same units and scored on the same 64 test contexts. The extension's ridge + residual-bootstrap baseline is refitted under each normalization.

Normalization	Ridge	White · MLP	White · S4	GP · MLP	GP · S4
Global	0.0933	0.1487	0.1778	0.2262	0.1702
History	0.0852	0.0855	0.0922	0.0886	0.0877

Fair CRPS, mean over three seeds. Pre-declared verdict: "C2 supported". The prior × head interaction shrinks from -0.085 to -0.008. 62.5% of the evaluated test contexts lie outside the central 95% of training history levels, so global scaling forces level extrapolation; history normalization removes it. With it, every network improves 1.7–2.6× and the best one ties ridge. The extension's S4 advantage was a normalization effect, not a head effect.

Kernels and window length

A pre-declared scan (docs/kernel_scan_protocol.md; 360 fits on four kernels — RBF, Matérn-5/2, Matérn-3/2, OU — and $F/\ell \in \{2, 4, 8\}$, three data draws) follows the assignment's own remedy for an empty task: lengthen the patch or shorten the lengthscale. With the GP source, S4/MLP CRPS lies between 0.978 and 1.004 in every cell. S4's residual-MSE disadvantage at the RBF anchor (1.18) disappears at $F/\ell \geq 4$, but no S4 advantage appears. Verdicts: residual MSE "Partial", CRPS "Refuted". For the Gauss–Markov kernels the pre-declared recommendation is "Keep TimeFlow's head".

What the assignment's metrics can see

From docs/final_report_addendum.md. **Residual group.** PCs 7–16 carry no structure: their eigenvalues match the normalized numerical nugget (5.6e-06), so "residual MSE" is mostly the PC3 error plus ten nugget directions. **Roughness.** True targets truncated to their first two PCs keep 0.88 of the target roughness, inside the ±20% band, although every residual PC is removed. Neither metric observes the Jacobian deficit of section 10; this is why the mechanism was invisible in the Task 2 table.

Summary of sections 10–11. The assignment's mechanism exists in the head Jacobian: with the matched GP source both heads leave the residual directions largely unlearned, and S4 recovers more of them than the MLP in every seed, but far from all. It is harmless in this cell, because those directions carry almost no variance in either the source or the target; under the GP source S4's CRPS advantage is only 2–3%. No kernel or window length gives S4 or whitening a material advantage. On ETTh1-OT, history normalization improves every network about twofold and brings them level with ridge, so normalization matters more than either the source or the head. Whitening learns the residual Jacobian but over-responds to perturbations and should be regularized before it is adopted.

A design recommendation with its limits

Recommendation. Do not change the latent-prior architecture on this stress test alone: the assignment's formal outcome remains inconclusive. Retain the existing small AdaLN MLP as the baseline, define a better discriminating success criterion prospectively, and investigate regularized PCA whitening next. Its lower measured errors make it a promising follow-up, not a demonstrated requirement for a structured prior. The all-fail branch requires withholding a method verdict; this study does not establish that TimeFlow needs S4, or that either pretrained system fails.

In the five-seed primary GP-source comparison, MLP residual MSE is 0.04927 versus 0.05827 for S4. Whitening reduces MLP residual MSE by **18.6%** and PC1 MSE by **8.8%**, while generated roughness remains close to the target. The paired residual-error difference for whitening is −0.00918, with a seed-bootstrap 95% interval of [−0.00961, −0.00855]. Roughness differences do not establish a separate whitening advantage. These intervals describe optimization variability on one fixed data draw; they are not uncertainty across GP families or real datasets.

The lower-learning-rate check matters: nine warm restarts improve validation objectives, so 6,000 updates cannot be claimed to establish convergence. Across the three restarted seeds, whitened MLP still has the lowest mean PC1 and residual errors, and S4 still has higher mean residual error than the ordinary MLP. The original checkpoints and five-seed results remain unchanged. Better coordinate weighting is a useful intervention here, but this does not prove a particular Jacobian or unused-dimension mechanism.

The shorter-lengthscale follow-up passes the geometry gate (GP velocity E_2=0.946), but again yields 0/15 literal successes and 15/15 roughness passes. GP-source residual MSE is 0.10859 for whitened MLP, 0.12372 for MLP and 0.13110 for S4. S4 now has slightly lower PC1 error than MLP, so head ordering depends on the metric. Shortening the lengthscale does not resolve the success-rule problem.

The literal two-sided raw 20% rule passes **0/25** primary runs, although **25/25** meet the roughness condition. Residual error is much smaller than PC1 error, rather than too large. The privileged analytic GP reference also fails that equality test. A secondary one-sided “residual no worse” reading would pass all 25, illustrating that raw error equality is not a reliable discriminator in this anisotropic geometry. Neither interpretation is substituted retrospectively for the declared primary flag.

Roughness agreement alone is insufficient: oracle Euler integration can suppress conditional variance while retaining near-correct adjacent jumps. The oracle also sees raw-history information unavailable to the networks. Normalized diagnostics and whitening depend on the declared eigenvalue floor; ten of sixteen GP directions are floored. We therefore withhold a general architectural or distributional-accuracy verdict. The optional real-series extension is omitted; further work should validate conditional calibration, additional data draws and tuning budgets before extending this recommendation.

Post-hoc analyses (sections 10–11). They leave the primary flags unchanged. The assignment's mechanism is present in the head Jacobian but harmless in this cell; no kernel or window length gives S4 a material advantage; and on ETTh1-OT history normalization matters more than either the source or the head.

Source and implementation references

Sundial / TimeFlow, ICML 2025; TSFlow, ICLR 2025; S4, ICLR 2022; Flow Matching, ICLR 2023. The two supplied assignment PDFs define the project’s required geometry, metrics, 20% rule and control.

Official code revisions: Sundial 3212e42564493f520593e5414af4367fc4b49226; state-spaces/S4 e757cef57d89e448c413de7325ed5601aceaac13; inspected TSFlow ff893176a41a1037f63efae674aab71480eee7d0. Apache-2.0 reference sources and licenses are preserved for Sundial and S4. TSFlow’s unlicensed repository source is inspected, not vendored. See vendor/manifest.json and docs/model_provenance.md for checksums and adaptations.

AI assistance was used for implementation, analysis tooling and writing. Numerical equivalence tests, independent known-value metric tests, split/geometry checks and saved experiment records provide verification; the report does not treat generated prose as evidence.